Model 1 on the v7 data: Vegetation–Water–Sand Dynamics

Rebuilt version, 2026-11-10

2026 CQU_MCM Problem B, Task 1

What this is. Model 1 rebuilt on the v7 data panel. It keeps the yearly state equation of `Reports/model_v7.tex` unchanged, adds one water relation, and fits only the two parameters the data can identify. All numbers come from `Code/v7_model1.py` and are stored in `Results/v7/model1/`. The detailed evidence is in `Docs/20261110_model1_v7_rebuild.md`.

1 The model

Vegetation (v7 equation, yearly step; V in % cover, v7 units):

$$V_{t+1} = V_t + \underbrace{r\,V_t\Big(1-\frac{V_t}{K}\Big)W_t}_{\text{growth}} - \underbrace{d\,V_t}_{\text{decay}} - \underbrace{k\,E_t\Big(1-\eta\frac{V_t}{K}\Big)V_t}_{\text{wind erosion}}. \tag{1}$$

Effective soil moisture (added relation):

$$W_t = \min\Big\{1,\ \frac{A_t}{c\,\min(1,\,2f_{c,t})\,\mathrm{ET}_{0g}}\Big\}, \qquad A_t = \eta_w I_t + G, \qquad f_{c,t} = \frac{V_t}{V_{\text{full}}}. \tag{2}$$

W is the water plants can use divided by what the canopy would transpire without stress. I is gross irrigation (mm/yr), G groundwater supply, $\mathrm{ET}_{0g} = \sum_m f(T_m)\,\mathrm{ET}_{0,m}$ the temperature-weighted reference evapotranspiration, f_c the ground-cover fraction. Root-zone storage (about 75 mm) is small against yearly fluxes of hundreds of mm, so at a yearly step moisture follows supply and demand and needs no storage equation.

Couplings. Water → vegetation: W scales growth. Vegetation → water: more cover raises demand and lowers W. Vegetation → sand: the shelter factor $(1-\eta V/K)$. Sand → vegetation: the erosion term.

Steady state. For given W the v7 closed form still holds:

$$x^* = \frac{V^*}{K} = \frac{rW - d - kE}{rW - kE\eta}, \qquad \text{and with } W=1,\ k=0: \quad x^* = 1 - \frac{d}{r} = 0.80. \tag{3}$$

Parameter	Value	Status
r	0.584 ± 0.180 yr^{-1}	fitted, pooled over 7 stations
d	0.117 ± 0.035 yr^{-1}	fitted, pooled over 7 stations
K	$1.15 \times Q_{0.95}$ of training-period V, per site; 38.1% on the section	v7 rule, unchanged
k	0 (estimate 0.00 ± 0.27)	not identified; scenario range 0–0.10
η	1	fixed; not identifiable
c	1.0	assumption, range 0.7–1.2 (FAO-56)
η_w	0.75	assumption, range 0.65–0.85
V_{full}	66.6% (v7 units)	from oasis pixels on the section
ET_{0g}	1163 mm/yr	v7 panel, stations YT and MF

2 Data

Data	Source	Used for
v7 monthly panel, 9 stations, 2015–2025	NASA POWER, MODIS MOD13Q1, ERA5 (`Data/v7/processed/`)	yearly V, E, ET_0; fitting r, d
217 pixels that went from bare sand to vegetation, 2003–2021	MODIS MOD13Q1 on the Yutian–Minfeng margin (`Data/v7/section_modis/`)	independent check of r, d

Yearly V is the mean of observed June–September months. Seven stations are vegetated (YT, MF, QM, RQ, KEL, YL, ALE); XT and TZ are desert. Target years to 2022 are used for fitting, 2023–2025 are held out.

3 What was checked in v7 and changed

	Finding	Evidence	Change
C1	Fitting four parameters per station on 8 yearly points does not work	v7 ablation table: median training R^2 0.004; 89% of fits end on a bound	one parameter set shared by all vegetated stations, K local
C2	k and η cannot be estimated	adding k: AIC $-282.4 \to -280.4$, $k = 0.00 \pm 0.27$; adding η: condition number 2.6×10^8	keep (r, d); k, η stay as scenario parameters
C3	The v7 water index (normalised GWETROOT) has no time signal	99.94% of its variance is between stations; yearly s.d. within a station 0.003–0.013; no irrigation in the reanalysis	W from the water relation; $W = 1$ at irrigated oasis stations in the fit
C4	$f(T)$ cannot be separated from r at a yearly step	yearly mean temperature varies by at most 1.8 °C within a station	$f(T)$ absorbed in r; kept inside ET_{0g}
C5	Imputed NDVI months were scored	9.3% of station-months imputed, 31% at KEL	only observed months enter yearly V
C6	Fixed end-members 0.05/0.85 understate ground cover	densest oasis pixels reach NDVI 0.583, i.e. 66.6% in v7 units	divide by V_{full} where a ground fraction is needed
C7	Task 2 used $K_s = 0.45 + 0.55\, I/60$ without basis	`v7_ablation.py`	replaced by the curve in Section 5

4 Data support

Stations, one year ahead (V/K)	Training RMSE	Hold-out RMSE	Hold-out R^2
Model (r, d)	0.054	0.068	0.40
Persistence (next year = this year)	0.060	0.058	–
Station mean	–	0.079	–

The model beats the station mean but not persistence on the hold-out years. The station series sit near their steady state, so they fix the ratio d/r and a relaxation rate, not much more.

Section pixels (R^2)	CON1–3 (K, V_0 fitted)	CON4 (never used)
Station r, d transferred, not refitted	0.40	0.31
All four parameters fitted freely	0.40	0.29
Mean baseline	0.00	−0.02

A free fit to the pixels gives $r = 0.64$, $d = 0.10$, against 0.58 and 0.12 from the stations. Two data sets with different sites, years and dynamics agree; this is the main support for r and d.

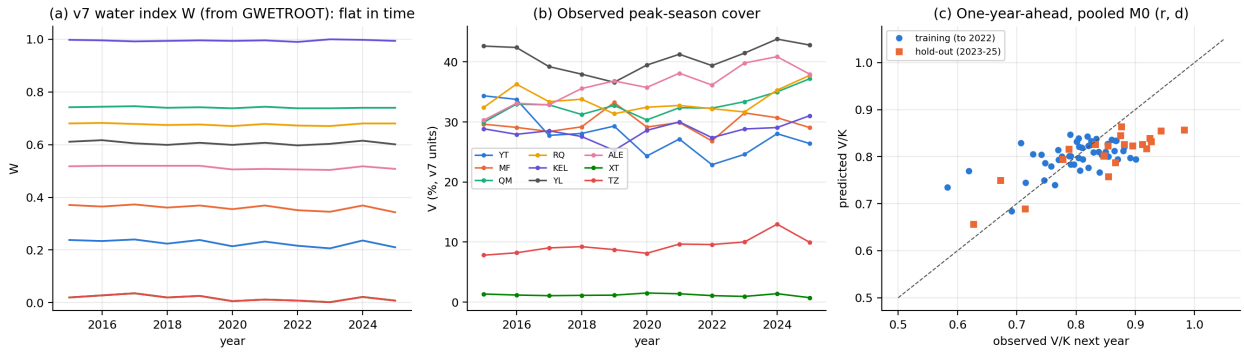


Figure 1: (a) The v7 water index is flat in time. (b) Observed peak-season cover. (c) One-year-ahead prediction of the pooled model.

5 Results

Gross irrigation (mm/yr)	0	100	200	300	400	600	800	1000
Steady V (%, v7 units)	0	8.3	13.7	17.4	20.1	23.9	26.3	28.1
Ground-cover fraction	0	0.13	0.21	0.26	0.30	0.36	0.40	0.42
W	0	0.26	0.31	0.37	0.43	0.54	0.65	0.77
Erosion relative to bare sand	1.00	0.78	0.64	0.54	0.47	0.37	0.31	0.26

- 1 mm/yr equals 10 m^3/ha/yr. Water stops limiting at about 1420 mm/yr.
- A new planting reaches 90% of its steady cover in 10–11 years.
- Rain only: cover goes to zero (XT station 1.2%; section desert pixels 0.2%).
- The natural ecotone (8.9% in v7 units) corresponds to about 81 mm/yr of groundwater supply.
- Steady cover at 300 mm/yr ranges from 14 to 21% across the sensitivity cases; r, d and c matter most.

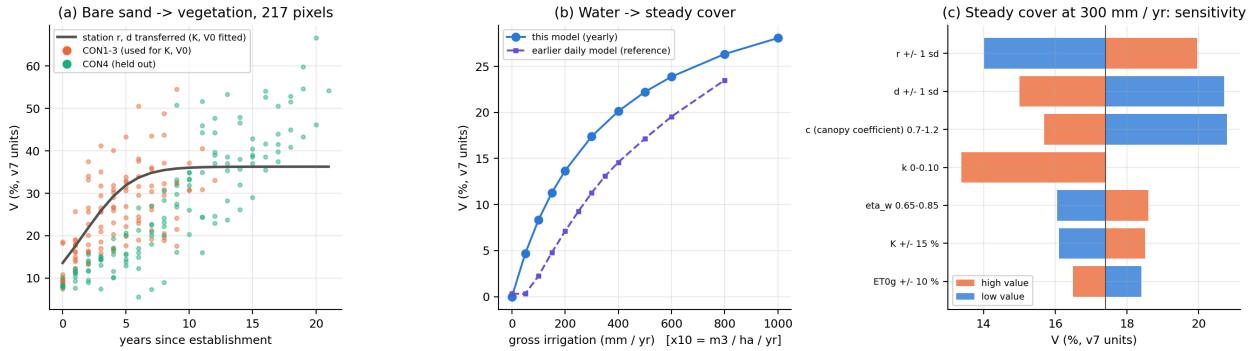


Figure 2: (a) Station parameters transferred to bare-to-vegetated pixels. (b) Water to steady cover, with the earlier daily model as reference. (c) Sensitivity of steady cover at 300 mm/yr.

6 Limits

1. One-year-ahead station cover is not predicted better than persistence.
2. The water-limited part of the curve has no measurement behind it; it depends on the assumed c and η_w.
3. When irrigation stops the model loses cover at rate d only (half-life 5.9 years). d was measured on watered vegetation, so real loss is probably faster; Task 3 should treat this as a scenario.
4. k is not identified; wind enters Task 3 as a scenario.
5. Stations are town coordinates inside oases and describe irrigated fringe vegetation, not shrub belts on dunes.